

Locating Multiple GPS Jammers Using Networked UAVs

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Abstract—Recent technologies, such as, Internet of Things and cloud services, increases the usage of small and low-cost networked unmanned aerial vehicles (UAVs), which needs to be robust against malicious global positioning system (GPS) attacks. Due to the availability of low-cost GPS jammers in the commercial market, there has been a rising risk of multiple jammers and not just one. However, it is challenging to locate multiple jammers because the traditional jammer localization via multilateration is applicable for only a single jammer case. Also, during a jamming attack, the positioning capability of an on-board GPS receiver is compromised given its inability to track GPS signals. We propose a simultaneous localization of multiple jammers and receivers (SLMR) algorithm by analyzing the variation in the front-end signal power recorded by the GPS receivers on-board a network of UAVs. Our algorithm not only locates multiple jammers but also utilizes these malicious sources as additional navigation signals for positioning the UAVs. We design a Gaussian mixture probability hypothesis density filter over a graph framework, which is optimized using a Levenberg–Marquardt minimizer. Using a simulated experimental setup, we validate the convergence and localization accuracy of our SLMR algorithm for various cases, including attacks with a single jammer, multiple jammers, and a varying number of jammers. We also demonstrate that our SLMR algorithm is able to simultaneously locate multiple jammers and UAVs, even for a larger transmitted power of the jammers.

Index Terms—Graph optimization, Internet of Things (IoT), localization, multiple jammers, probability hypothesis density (PHD), unmanned aerial vehicles (UAVs).

LIST OF SYMBOLS

α_k	Random variable of the Gaussian distribution Γ_k .
β_k	True location and transmit power of the k th jammer.
ΔT	Update time interval of the GM PHD filter.
δt	Integration time interval in GPS tracking.
$\dot{\Sigma}_t$	Covariance of the rate of the relative distances between each jammer–receiver pair.
ϵ	Cut-off threshold considered for the $d(\cdot, \cdot)$ in $d_\epsilon(\cdot, \cdot)$.
ϵ'	Empirical threshold to penalize $\lambda_2(\cdot, \cdot)$.
η_{ik}	Relative distances between k th jammer and i th UAV.

Γ_k	Multivariate Gaussian distribution of the 3-D position and transmit power of the k th jammer.
ι_{ik}	Noise related to the k th jammer and i th UAV.
$\lambda_1(\cdot, \cdot)$	Minimum data association error between estimated and true jammers; first component of the OSPA metric.
$\lambda_2(\cdot, \cdot)$	Error between the estimated and actual number of jammers; second component of the OSPA metric.
Λ_t	RFSs relative distance between each jammer–UAV pair.
$\dot{\mathbf{x}}_{i,t}$	3-D velocity of the i th UAV.
$\dot{\mathbf{y}}_{k,t}$	3-D velocity of the k th jammer.
$\mathbf{e}_t$	Error function at the t th time instant.
$\mathbf{x}_i$	3-D position of the i th UAV.
$\mathbf{y}_k$	3-D position and transmit power of the k th jammer.
$\nu(\cdot)$	Posterior intensity of the elements in the RFS state.
$\omega\{\cdot\}$	Thermal noise of the receiver.
$\Omega_{i,t}$	Covariance of the i th UAV position at the t th time.
Φ_k	Transmit power of the k th jamming source.
σ_k^2	Receiver noise associated with the k th jammer.
Σ_t	Covariance of the relative distances between each jammer–receiver pair.
τ	Element from sampling times $[0, (1/f_s), (2/f_s), \dots, \delta t - (1/f_s)]$.
Θ_t	RFS transmit power of the jammers.
ζ_k	Neyman–Pearson threshold of the k th jammer.
A_0	Null hypothesis that the k th jammer is not detected.
A_1	Hypothesis that the k th jammer is detected.
b	Number of recent time instants used in optimization.
c	Speed of light = 299 792 458 m/s.
$d_\epsilon(\mathbf{u}, \mathbf{v})$	Distance between $\mathbf{u}$ and $\mathbf{v}$ cut off at $\epsilon > 0$.
D_t^r	Coefficients of the Fisher information (FI) matrix, where $r = 11, 12, 21, 22$.
e_{LB}	Lower bound on the OSPA metric.
e_{OSPA}	Optimal subpattern assignment (OSPA) metric.
F	State transition matrix of the UAV motion model.
f_s	Sampling frequency of the GPS front-end.
f_{L1}	Frequency of GPS L1 C/A signals, 1575.42 MHz.
$g(\cdot, \cdot)$	Motion model of the UAVs.
G_{tx}	Antenna gain of the receiver.
G_{tx}	Antenna gain of the transmitter.
$h(\cdot, \cdot)$	Measurement model of k th jammer and i th UAV.
$H_{i,t}^j$	Linearized measurement model of i th UAV using j th Gaussian component at t th time.
J	Number of Gaussian components in the GM-PHD.
$J_{k,t}$	FI matrix for the k th jammer at t th time.

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$J_{\max}$	Maximum acceptable components in GM-PHD filter.
L	Number of UAVs in our SLMR algorithm.
m^j	Mean of the j th Gaussian component of the posterior intensity in the GM-PHD filter.
$M_{\text{true},t}$	Actual number of jammers affecting UAVs at t th time.
M_t	Number of GM-PHD estimated jammers.
N	Number of satellites in view.
P^j	Covariance of j th Gaussian component of the posterior intensity in the GM-PHD filter.
$p_0(\cdot)$	Conditional probability under the null hypothesis A_0 .
$p_1(\cdot)$	Conditional probability under hypothesis A_1 .
$p_{D,t}$	Probability of detection given the jammer state $\mathbf{y}$.
$p_{i,\text{init}}$	Average power during initialization at i th UAV.
$p_{i,t}$	Average GPS front-end power at i th UAV and t th time.
p_S	Survival probability of the state $\mathbf{y}$ at the $(t + 1)$ th time.
Q_t	Process noise covariance matrix.
R_t	Measurement noise covariance matrix.
$s^n\{\cdot\}$	GPS satellite signal from the n th satellite.
$s_t\{\cdot\}$	Received GPS signal at t th time instant.
T	Weight threshold in the GM-PHD measurement update.
$u_{i,t}$	Control input from the motion model to the i th UAV.
W	Number of measurements stored during initialization.
w^j	Weight of the j th Gaussian component of the posterior intensity in the GM-PHD filter.
$Y_{\text{true},t}$	RFS true state of the nonlinear PHD filter.
Y_t	RFS state of the nonlinear GM-PHD filter.
z_i	Change in the average front-end power at the i th UAV.
Z_t	RFS measurement at the t th time instant.

I. INTRODUCTION

DUE TO the rapid advances in the Internet of Things (IoT) technologies [1], the use of small and low-cost unmanned aerial vehicles (UAVs) [2] is becoming predominant in commercial, research and military fields. IoT provides a robust platform for the UAVs to be deployed to remote locations, easily navigated, reprogrammed, and monitored in real-time. A few commercial applications of UAVs over IoT include transportation [3], geo-surveying [4], and Internet delivery [5]. In addition, companies across the world have dedicated ongoing commercial UAV projects, such as Amazon Prime Air [6], Project Wing [7], Aquila UAV, and so on.

UAVs play a crucial role during various human-inoperable monitoring and rescue operations [8], such as forest fires [9], and earthquakes and accident site inspections [10]. Similarly, on a global scale, the North American Treaty Organization utilizes military UAVs, such as Global Hawk [11], for protection, maritime security, and assistance [12]. The number of UAVs registered with Federal Aviation Administration

exceeded 300 000 in January 2016 [13]. Therefore, ensuring reliable UAV navigation is of utmost importance for their safe deployment. Global positioning system (GPS), which provides absolute position and works under all visual and weather conditions, is at the heart of many state-of-the-art navigation systems [14]. GPS has global coverage, is free to all users and provides microsecond-level timing and meter-level positioning accuracy [15].

A. Networked Localization of UAVs via IoT and Related Work

For navigation, the constraints on computation and payload capacity cause most of the small and low-cost commercial UAVs to be mounted with only GPS and inertial measurement units (IMUs) [16]. However, the IMUs on-board small UAVs are usually of low cost and low performance. Therefore, in this paper, we describe our localization framework [17] using GPS, which can be extended to incorporate other sensors, such as IMUs, ultrawide band (UWB), and so on.

Cooperative localization [18] is emerging as a promising framework to ensure robust GPS navigation. It functions as a *distributed brain* to gather additional information about the unknown environment [19]. Networked UAVs utilize the geographical diversity of multiple UAVs and their individual sensor measurements to execute cooperative localization.

IoT efficiently connects multiple UAVs into a ubiquitous network [20] with each UAV representing a node in the network. The easy availability of WiFi and cellular connection [21] establishes inter-UAV communications [22]. Similarly, the use of cloud services [23], such as Amazon Web Services, Google cloud platform (GCP) enable collective computational processing of measurements gathered from multiple UAVs. This also minimizes the requirement for large computational resources on-board the UAVs.

Prior research on the localization of networked UAVs includes augmenting the GPS with other sensors, such as IMU, camera, LiDAR via dead-reckoning [24], or filtering techniques [25]. Multiple UAVs are localized using extended Kalman filtering [26] and decentralized cooperative localization algorithms [27] based on the inter-UAV ranges obtained via UWBs. Some researchers exploit the known landmarks [28], such as the GPS reference stations and beacons in the vicinity to localize the networked UAVs using simultaneous localization and mapping (SLAM) [29] techniques.

B. Multiple Jammers and Related Work on Jammer Localization

Most of the proposed cooperative algorithms for the localization of networked UAVs assume normal GPS operating conditions with errors occurring due to ionospheric scintillation, urban multipath, terrain masking, satellite blockage, and so on. However, due to low signal power, the GPS signals are vulnerable to jamming attacks [30] and [31]. A jammer broadcasts high power signals in the GPS frequency band and causes the receiver to lose track of satellites [32], due to which the navigation solution can no longer be computed by the receivers.

A few notable, real-world jamming incidents in the past occurred near the Newark airport [33] and in the U.K. [34], where the GPS-based navigation systems malfunctioned for a significant time period. Nowadays, low-cost jammers are becoming easily available in the commercial market [35]. There is an increasing risk of imminent threats from multiple jammers and not just one [36]. Therefore, it is critical to account for the presence of multiple jammers to ensure the prompt neutralization of all these malicious devices. In addition, given the inability of the receivers to track GPS signals during jamming, we need alternative techniques to localize their position and facilitate continuous operation.

Researchers [37] and [38] have proposed an efficient jamming localization algorithm by rotating the UAV at multiple fixed positions to create the antenna gain pattern, estimate the jamming signal strength and its bearing. While the algorithm demonstrates compelling merits for stationary infrastructure, there are practical limitations related to the implementation, such as high costs and long delays. Previous papers [39] and [40] perform cross-correlation across the receivers to estimate the time difference of arrival (TDOA), which is used in localizing the jammer. Similarly, another work [41] implements maximum likelihood estimation to locate the jammer based on the difference in the received signal strength (RSS) measurements across a network of receivers with known positions.

Prior research on jammer localization is based on multilateration for which the measurements are generated using TDOA, directional antenna, or RSS. Despite the strong and straightforward proof-of-concept, the existing techniques are efficient for a single jammer case. This is because in the presence of multiple jammers, they fail to estimate the number of jammers and distinguish the signals received from different jammers. Even though the aforementioned techniques separately address certain aspects of cooperative localization of networked UAVs and GPS jammer localization, to our knowledge, none of the existing works simultaneously localize both networked UAVs and multiple jamming sources.

C. Our Contributions

In our prior work [42], we discussed the preliminary framework of our proposed simultaneous localization of multiple jammers and receivers (SLMR) algorithm that utilizes the variation in the received GPS front-end signal power to simultaneously locate multiple jammers and multiple UAVs. We utilize the jammers as additional navigation signals to constrain the positioning errors of a network of UAVs. In this paper, we present our new contributions as follows.

- 1) We develop a graphical framework constrained by the UAV motion model and the nonlinear Gaussian mixture-probability hypothesis density (GM-PHD) filter [43]. We optimize the graph using the Levenberg-Marquardt (LM) minimizer [44] to simultaneously locate multiple jammers and multiple UAVs. As compared with our prior work [42], we remove the assumption that each jammer transmits the same power by modeling the transmitted powers as additional unknowns to be estimated using a GM-PHD filter.

- 2) Theoretically, we compute the lower error bound on the localization of multiple jammers via optimal subpattern assignment (OSPA) [45] in two stages. First, we formulate the posterior Cramér Rao bound (PCRB) [46] to analyze the recursive GM-PHD filter. Second, we characterize the probability of jamming detection and false alarm using a likelihood ratio test via Neyman–Pearson decision rule [47].
- 3) We validate the localization accuracy and convergence of our SLMR algorithm by conducting extensive simulated experiments, such as attacks with a single jammer, multiple jammers, a varying number of jammers, and also for a larger transmitted power of the jammers.

The rest of this paper is organized as follows. Section II describes the high-level architecture and the key characteristics of our algorithm. Section III explains our SLMR algorithm in detail. Section IV derives the theoretical lower error bound on the localization of multiple jammers. Section V experimentally validates our algorithm by considering a network of UAVs subjected to various simulated cases of multiple jammers. Section VI concludes this paper.

II. SIMULTANEOUS LOCALIZATION OF MULTIPLE JAMMERS AND RECEIVERS

In this section, we describe the architecture of our proposed SLMR algorithm. We consider a network of L geographically distributed UAVs, each with an on-board GPS receiver that is affected by multiple GPS jammers.

A. GPS and Jammer Signal Model

The received GPS signal $s_t\{\tau\}$ is a superposition of satellite signals $s^n\{\tau\}$ from N satellites [48], which is represented as

$$s_t\{\tau\} \triangleq \sum_n^N s^n\{\tau\} + \omega\{\tau\} \quad (1)$$

where τ is an element from a vector of sampling times $[0, (1/f_s), (2/f_s), \dots, \delta t - (1/f_s)]$ within a time interval δt and a sampling frequency of f_s . $\omega\{\tau\}$ is the thermal noise of the receiver, mostly additive Gaussian.

Next, we denote $M_{\text{true},t}$ as the total number of jammers affecting a network of UAVs at the t th time instant. We assume an unobstructed direct line-of-sight signal between each jammer–receiver pair. Using the principles of free space path loss [49], we model the effect of the k th jammer on the i th UAV. The received jamming power p at the GPS front-end, which is located at a distance η from a jammer is given by

$$p = \Phi G_{tx} G_{rx} \left(\frac{c}{4\pi\eta f_{L1}} \right)^2 \quad (2)$$

where Φ denotes the transmit power of the jamming source and G_{tx} and G_{rx} denote the antenna gain of the transmitter and receiver. The speed of light is given by $c = 299\,792\,458$ m/s and the frequency of $L1$ C/A GPS signals is $f_{L1} = 1575.42$ MHz. In this paper, we do not focus on jamming detection and consider that the jamming attacks are detected using existing literature [50], [51].

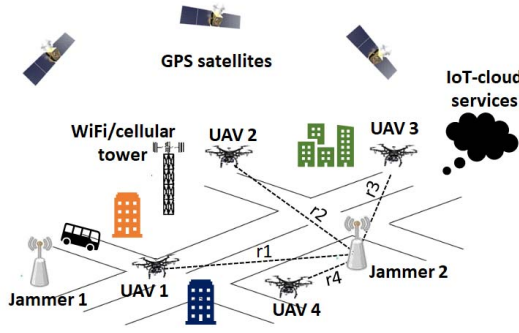


Fig. 1. Proposed SLMR algorithm utilizes the sequential variation in the received front-end GPS power and the motion model of a network of UAVs to localize multiple jammers.

B. Aspects of Our Algorithm

We utilize three available resources in our proposed SLMR algorithm, which are explained as follows.

1) *Variation in the Received Front-End GPS Signal Power:* Under jamming, based on the proximity to the attacker, the total received power of the GPS signal at the front-end increases. Therefore, we leverage the sequential variation in the received GPS front-end signal power across a network of L geographically separated UAVs to execute our SLMR algorithm, as seen in Fig. 1. We assume that the receivers do not enable dynamic automatic gain control [52] so that the changes in the received power are detected. We utilize the IoT-based wireless communication and cloud services to process the measurements obtained from a network of receivers on-board the UAV in a centralized manner.

2) *PHD Filter:* We formulate a PHD filter [53] to estimate the unknown number of jammers and the data association among each jammer–receiver pair. Unlike the conventional multitarget tracking [53], which propagates the probability density of the states, PHD filter propagates the first order statistical moment, i.e., posterior intensity of the states, which makes it less computationally expensive. In a PHD filter, we model the states and measurements as random finite sets (RFSs) [43], in contrast to the Kalman filter, which models them as a fixed-length vector [54].

For an RFS measurement Y on χ with probability distribution p , its intensity ν on χ , for each region $S \subseteq Y$ [43]

$$\int |Y \cap S| p(dY) = \int_S \nu(\mathbf{y}) d\mathbf{y}.$$

The RFS measurement Y is comprised of the received front-end power values observed at a network of UAVs. Similarly, the RFS state represents the 3-D positions and transmit powers of multiple jammers, and the cardinality $|\cdot|$ of the RFS state is modeled as a random variable, which indicates the number of jammers. Intuitively, the integral of $\nu(\cdot)$ over any region S gives the expected number of elements of Y that are in S . Therefore, the local maxima of the intensity $\nu(\cdot)$ are points in χ with the highest local concentration of the expected number of elements, which are later used to generate the elements of Y . Therefore, given multiple jammers, a multimodal measurement distribution is tracked by the PHD filter, as seen in

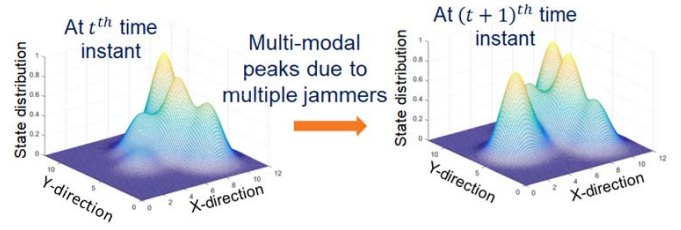


Fig. 2. Multimodal measurement distribution, observed due to multiple jammers, is tracked by the GM-PHD filter.

Fig. 2, where the coordinates of different peaks indicate the characteristics of different jammers.

3) *Graph-SLAM:* Our graphical framework [55] is inspired by a well-known technique in robotics, known as SLAM [56]. SLAM utilizes the sensor measurements to estimate the features in an unknown 3-D map while simultaneously localizing the robot within it. Analogous to the SLAM approach in robotics, we design a graph-SLAM framework, where the *robots* are our UAVs, and the *features* represent the jamming sources.

Our graph is constrained by the UAV motion model and its corresponding received front-end GPS signal power to simultaneously localize the jammers and UAVs. During jamming attacks, when the GPS navigation solution of UAVs can no longer be computed using conventional scalar tracking, graph optimization corrects for the drift errors, which otherwise get accumulated via dead-reckoning.

C. SLMR Architecture

Based on the various aspects discussed in Section II-B, we outline our SLMR architecture, shown in Fig. 3, as follows.

- 1) Once jamming is detected, we initialize our SLMR algorithm using a network of L UAVs and obtain the received front-end signal power from the GPS receivers on-board the UAVs.
- 2) We execute the measurement update of our GM-PHD filter to compute the posterior intensity of the 3-D positions and transmit powers of multiple jammers. We then estimate the number of jammers M_t , the RFS relative distance between jammers and UAVs Λ_t and the RFS transmit power Θ_t .
- 3) Later, to constrain the localization errors, we optimize the graph on a couple of the most recent time instants. The error function incorporates the constraints, obtained due to the UAV motion model and the GM-PHD filter. We perform graph optimization using the LM algorithm to compute the position and velocity of all the UAVs as well as the multiple jammers.
- 4) Next, based on the predicted position of the UAVs and the jammers for the next time instant, we iterate the above steps to converge to the accurate number of jammers and simultaneously localize multiple jammers and a network of UAVs.

III. SLMR ALGORITHM

In this section, we explain our SLMR algorithm and also describe the corresponding theoretical formulation in detail.

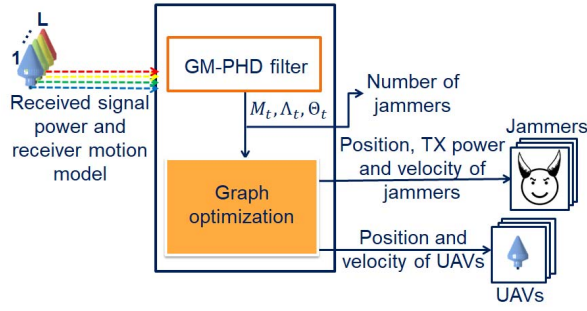


Fig. 3. Architecture of the SLMR algorithm. SLMR processes the received front-end GPS signal power and the motion model via a GM-PHD filter and graph optimization to simultaneously localize the multiple jammers and the UAVs. The output from the GM-PHD filter are the number of jammers M_t , RFS relative distance Λ_t , and RFS transmit power Θ_t .

A. Change in Received GPS Front-End Power

In Fig. 4, we plot the variation in the average received power for a stationary and moving UAV by blue and red, respectively, during the presence and absence of jamming. For i th satellite, we calculate the change in averaged received power z_i , which represents the jammer power described in (2), as

$$z_{i,t} = p_i - p_{i,\text{init}} = \sum_k \left(\frac{\Phi_k}{\eta_{ik}^2} \right) + \iota_{ik} = \sum_k h(\mathbf{x}_i, \mathbf{y}_k) \quad (3)$$

where p_i represents the average received GPS front-end signal power at the i th UAV and $p_{i,\text{init}}$ represents the average front-end power received at the i th UAV during initialization, which is explained later in Section III-F. Also, $\mathbf{x}_i = [x, y, z]_i$ denotes the position of the i th UAV, $\mathbf{y}_k = [x, y, z, \phi]_k$ represents the 3-D position and transmit power of the k th jammer, such that $\Phi_k = \mathbf{y}_k[\phi]$ denotes the transmit power of the k th jammer, $\eta_{ik} = \|\mathbf{x}_i - \mathbf{y}_k[x, y, z]\|$ denotes the relative distance between the i th UAV, and the k th jammer and ι_{ik} denotes the k th jammer and i th UAV specific errors. $h(\cdot, \cdot)$ denotes the measurement model. $Z_t = \{z_1, \dots, z_i, \dots, z_L\}$ represents the RFS measurement provided to the measurement update of the PHD filter [57].

We utilize the available WiFi signals to establish a communication between each UAV and a centralized IoT-based cloud platform [23]. The change in the average front-end GPS signal power Z_t measured at a network of UAVs and control input $u_{i,t} \forall i \in \{1, \dots, L\}$ obtained from the UAV motion models are sent to the cloud where the subsequent processing takes place. Thereafter, the SLMR estimated position and velocity is sent back to the respective UAVs and the positions of jamming sources are communicated to the UAVs as well as authorities. We extended our prior work [58] on single UAV-based communication setup using Robot Operating System [59] to multiple UAVs for this paper.

B. Graphical Framework

Considering the motion of UAVs to be independent of each other, we formulate a bipartite graph to simultaneously localize the jammers and UAVs using sequential data. The entire graph framework of our SLMR algorithm depicting L UAVs and

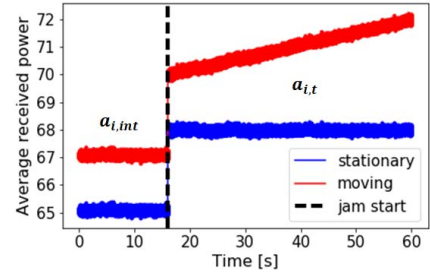


Fig. 4. Example of the average received power at a stationary UAV denoted by the blue line and a moving UAV denoted by the red line. After jamming starts, there is a significant jump in the received power. The motion of UAVs also vary the average received power, as indicated by the red line during 15–60 s. The variation in these measurements z_i is analyzed by our SLMR algorithm.

M_t jammers are represented in terms of two subgraphs: one is across time for the i th UAV, as seen in Fig. 5(a) and the other is across the UAVs for the t th time instant, as shown in Fig. 5(b). Based on Fig. 5(a), the various nodes in our graph framework are explained as follows.

- 1) The blue nodes depict the sequential time series of the 3-D position $\mathbf{x}_{i,1:t}$ and velocity $\dot{\mathbf{x}}_{i,1:t}$ of the i th UAV.
- 2) The orange nodes in the bottom layer represent the sequential data of 3-D position and transmit power $\mathbf{y}_{1:M_t,1:t}$ as well as the velocity of jammers $\dot{\mathbf{y}}_{1:M_t,1:t}$.
- 3) The orange nodes in the top layer denoted by $u_{i,1:t}$ represent the sequential measurements obtained from the motion model of the i th UAV.

The nodes of the different jammers are not interconnected as we assume that the jammers are not dependent on each other. At any time instant, the edges connecting the blue nodes to the orange nodes, as seen in Fig. 5(b), represent the constraint η_{ik} obtained from the measurement update of the PHD filter, as explained later in Section III-C. We perform graph optimization to estimate the position and velocity of the UAVs $\mathbf{x}_{1:L,1:t}, \dot{\mathbf{x}}_{1:L,1:t}$ as well as the multiple jammers $\mathbf{y}_{1:M_t,1:t}, \dot{\mathbf{y}}_{1:M_t,1:t}$.

C. Measurement Update of the GM-PHD Filter

We estimate the number of jammers as well as the data association across UAVs and jammers by employing a nonlinear GM-PHD recursion. The nonlinearity in the measurement model $h(\cdot, \cdot)$, given by (3), is explained in Section III-A. Based on the current estimate of the number of jammers M_t , the RFS state Y_t representing the 3-D position of jammers and their transmit powers is defined as

$$Y_t = \{\mathbf{y}_k, k \in \{1, \dots, M_t\}\}. \quad (4)$$

We approximate the first order statistical moment as a sum of the weighted Gaussian densities $\mathcal{N}(\cdot)$, which represents a multimodal distribution, as described in Fig. 2. By approximating the posterior intensity as a GM, the GM-PHD filter converges to a closed form solution [43]. Therefore, by dropping the subscript, we model the posterior intensity $\nu(\mathbf{y})$, of

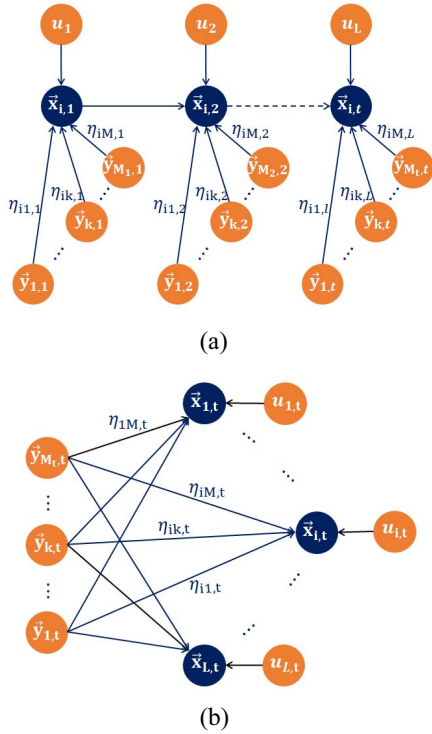


Fig. 5. Graphical framework of the SLMR algorithm. The orange nodes correspond to the UAV motion model $u_{i,t}$ and the jammer navigation solution $y_{1:M,t}$. Blue nodes represent the sequential navigation solution $\mathbf{x}_{i,1:t}$ of the i th UAV. (a) Sequential data for the i th UAV and (b) L UAVs at t th time instant, constrained via the received GPS signal power and UAV motion model. The edges are constrained by the relative distances η_{ik} estimated via the measurement update of the GM-PHD filter.

any element $\mathbf{y} \in Y_t$ in the RFS state as

$$v(\mathbf{y}) = \sum_{j=1} w^j \mathcal{N}(\mathbf{y}; m^j, P^j) \quad (5)$$

where $\mathcal{N}(\cdot)$ indicates the Gaussian density of the element $\mathbf{y}$, m^j denotes the mean of j th component of the GM, which represents the positions and transmit powers of a jammer, P^j denotes its corresponding covariance, and w^j represents the GM weights associated with the j th component. In this paper, the subscript t indicates the updated variable estimated during the measurement update whereas the subscripts $t|t-1$ and $t+1|t$ indicates the predicted variables estimated during the time update.

Based on the predicted estimate of the UAVs $\mathbf{x}_{1:L,t|t-1}$ obtained via a motion model $g(\cdot, \cdot)$ and the variation in the received signal power across a network of L UAVs Z_t , we compute the measurement update of the posterior intensity $v_t(\mathbf{y})$ for any state $\mathbf{y}$ as follows:

$$v_t(\mathbf{y}) = (1 - p_{D,t})v_{t|t-1}(\mathbf{y}) + \sum_{z_i \in Z_t} v_{D,t}(\mathbf{y}; z_i) \quad (6)$$

$$v_{D,t}(\mathbf{y}; z_i) = \sum_{j=1}^{J_{t|t-1}} w_t^j(z_i) \mathcal{N}(\mathbf{y}; m_{t|t}^j(z_i), P_{t|t}^j(z_i)).$$

In (6), $p_{D,t}$ denotes the probability of detection given the state of the jammer $\mathbf{y}$. $v_{D,t}(\mathbf{y}; z_i)$ denotes the detected component of the posterior intensity at the t th time

instant based on the received measurements $z_i \in Z_t$ with the following associated mean $m_{t|t}^j$ and covariance $P_{t|t}^j$, respectively. Similarly, $(1 - p_{D,t})v_{t|t-1}(\mathbf{y})$ represents the misdetection component based on the predicted posterior intensity $v_{t|t-1}(\mathbf{y})$. $v_{D,t}(\mathbf{y}; z_i)$ is represented as a GM consisting of $J_{t|t-1}$ components and the following associated parameters: $w_t^j(z_i)$ denotes the weights allocated to j th Gaussian density corresponding to each measurement z_i

$$w_t^j(z_i) = \frac{p_{D,t} w_{t|t-1}^j(z_i) q_t^j(z_i)}{\kappa_t(z_i) + p_{D,t} \sum_{l=1}^{J_{t|t-1}} w_{t|t-1}^l(z_i) q_t^l(z_i)}$$

$$q_t^j(z_i) = \mathcal{N}(z_i; m_{t|t}^j(z_i), (R_t + H_{i,t} P_{t|t-1}^j H_{i,t}^T))$$

$$H_{i,t}^j = \frac{\partial h(\mathbf{x}_{i,t|t-1}, \mathbf{y})}{\partial \mathbf{y}} \Big|_{\mathbf{y}=m_{t|t-1}^j}$$

$$K_t^j = P_{t|t-1}^j H_{i,t}^T (R_t + H_{i,t} P_{t|t-1}^j H_{i,t}^T)^{-1}$$

$$m_{t|t}^j(z_i) = m_{t|t-1}^j + K_t^j (z_i - H_{i,t}^j m_{t|t-1}^j)$$

$$P_{t|t}^j = (I - K_t^j H_{i,t}^j) P_{t|t-1}^j. \quad (7)$$

In (7), the mixture weights $w_t^j(z_i)$ are updated using $q_t^j(z_i)$, which indicates the j th Gaussian density of the measurement z_i based on the updated estimate of the mean $m_{t|t}^j$, measurement noise covariance R_t and covariance $P_{t|t-1}^j$. $\kappa_t(z_i)$ represents the intensity of clutter associated with measurement z_i , initialized as a Poisson RFS based on the specifications of the GPS front-end sensor. To calculate $q_t^j(z_i)$, we also utilize R_t , which represents the measurement noise covariance matrix and $H_{i,t}^j$, which represents the measurement model $h(\cdot, \cdot)$ of the i th UAV linearized using the j th GM component at the t th time instant.

After executing (6) and (7), the total number of Gaussian components increase from $J_{t|t-1}$ to $J_{t|t} = J_{t|t-1} + L J_{t|t-1}$. We express the updated posterior intensity as a GM with J_t number of components as

$$v_t(\mathbf{y}) = \sum_{j=1}^{J_{t|t}} w_t^j \mathcal{N}(\mathbf{y}; m_{t|t}^j, P_{t|t}^j). \quad (8)$$

Thereafter, we prune the Gaussian components by fixing the upper bound on the number of Gaussian components $J_{\max}$ and also ignoring the ones with weights less than a threshold $w_t^j < T$. Also, we merge the ones with similar Gaussian distribution characteristics so as to improve the computational efficiency. After the measurement update, we compute the number of jammers M_t , the RFS state Y_t which equals the mean $m_{t|t}^j$ of the Gaussian components, the RFS relative distance Λ_t , and RFS transmit power Θ_t , as

$$M_t = (1 - p_{D,t}) M_{t|t-1} + \sum_{z_i \in Z_t} \sum_{j=1}^{J_{t|t-1}} w_t^j(z_i)$$

$$Y_t = \{y_{k,t} = m_{t|t}^j, j \in \{1, \dots, J_t\} \forall m_{t|t}^j > T\}$$

$$\Theta_t = \{\Phi_{k,t} = \mathbf{y}[\phi], \mathbf{y} \in Y_t\}$$

$$\Lambda_t = \{\eta_{ik,t} = \|\mathbf{x}_{i,t|t-1} - \mathbf{y}[x, y, z]\|, i \in \{1, \dots, L\}, \mathbf{y} \in Y_t\}. \quad (9)$$

D. Graph Optimization

We incorporate the constraints obtained from the motion model and the measurement update of the GM-PHD filter Λ_t to our existing graph framework. The motion model is obtained from either the known UAV dynamics or sensors, such as IMU, camera and so on. We formulate the error function as

$$\begin{aligned} \mathbf{e}_t(\mathbf{x}_{1:L}, \dot{\mathbf{x}}_{1:L}, \mathbf{y}_{1:M_t}, \dot{\mathbf{y}}_{1:M_t}) \\ = \sum_i \sum_k \left(\frac{\Phi_{k,t}}{\eta_{ik,t}^2} - h(\mathbf{x}_i, \mathbf{y}_k) \right)^T \Sigma_t^{-1} \left(\frac{\Phi_{k,t}}{\eta_{ik,t}^2} - h(\mathbf{x}_i, \mathbf{y}_k) \right) \\ + \sum_i \sum_k (\dot{\eta}_{ik,t} - (\dot{\mathbf{x}}_i - \dot{\mathbf{y}}_k))^T \dot{\Sigma}_t^{-1} (\dot{\eta}_{ik,t} - (\dot{\mathbf{x}}_i - \dot{\mathbf{y}}_k)) \\ + \sum_i ([\mathbf{x}_i, \dot{\mathbf{x}}_i]^T - g(u_t, \mathbf{x}_{i,t-1|t-1}))^T \Omega_{i,t}^{-1} \\ \times ([\mathbf{x}_i, \dot{\mathbf{x}}_i]^T - g(u_t, \mathbf{x}_{i,t-1|t-1})) \end{aligned} \quad (10)$$

where u_t denotes the control input given to the motion model $g(\cdot, \cdot)$ and $\Omega_{i,t}^{-1}$ denote the covariance of the i th UAV position at the t th time instant. The objective function $\mathbf{e}_t$ consists of three components [57]. The first component compares the constraints obtained from the update step of our nonlinear GM-PHD filter, i.e., $\Phi_k \in \Theta_t$, $\eta_{ik} \in \Lambda_t$ to that of the expected measurement model $h(\cdot, \cdot)$ summed across each jammer–receiver pair. The second component represents the error related to the relative velocity measurement $\dot{\eta}_{ik,t} = [(\eta_{ik,t} - \eta_{ik,t-1})/\Delta T]$ obtained from the relative distances at the t th and $(t-1)$ th time instant, where ΔT denotes the update time interval of the GM-PHD filter. In addition, Σ_t and $\dot{\Sigma}_t$ denotes the covariance of the relative distances and relative rates between each jammer–receiver pair. Finally, we compare the unknown position and velocity of the receiver to be estimated with that of the expected position of the receiver based on motion model $g(\cdot, \cdot)$. Thereafter, we carry out the optimization procedure using an LM minimizer.

We execute graph optimization across the most recent b time stamps to constrain the errors of UAVs and jammers. This is executed by minimizing the total error function given by

$$\begin{aligned} (\mathbf{x}_{1:L,t-b:t}, \dot{\mathbf{x}}_{1:L,t-b:t}, \mathbf{y}_{1:M_t,t-b:t}[x, y, z], \dot{\mathbf{y}}_{1:L,t-b:t}) \\ = \operatorname{argmin} \sum_{n=t-b}^t \mathbf{e}_n(\mathbf{x}_{1:L}, \dot{\mathbf{x}}_{1:L}, \mathbf{y}_{1:M_t}, \dot{\mathbf{y}}_{1:M_t}). \end{aligned} \quad (11)$$

Next, based on the current estimates of the position and velocity of the UAVs $\mathbf{x}_{1:L,t}, \dot{\mathbf{x}}_{1:L,t}$ and the jammers $\mathbf{y}_{1:M_t,t}[x, y, z], \dot{\mathbf{y}}_{1:L,t}$, we correct the RFS state Y_t , the RFS relative distance Λ_t and the RFS transmit power Θ_t . This is later provided to the time update step of the GM-PHD filter.

E. Time Update of the GM-PHD Filter

Lastly, we compute the predicted intensity for the next instant $t+1$ as

$$\begin{aligned} v_{t+1|t}(\mathbf{y}) &= v_{S,t+1|t}(\mathbf{y}) + \gamma_{t+1}(\mathbf{y}) \\ v_{S,t+1|t}(\mathbf{y}) &= p_{S,t} \sum_{j=1}^{J_t} w_t^j \mathcal{N}(\mathbf{y}; m_{S,t+1|t}^j, P_{S,t+1|t}^j) \end{aligned}$$

$$\begin{aligned} \gamma_{t+1}(\mathbf{y}) &= \sum_{j=1}^{J_{\gamma,t+1}} w_{t+1}^j \mathcal{N}(\mathbf{y}; m_{\gamma,t+1}^j, P_{\gamma,t+1}^j) \\ M_{t+1|t} &= p_{S,t} M_t + \frac{1}{L} \sum_{j=1}^{J_{\gamma,t}} w_{\gamma,t+1}^j \end{aligned} \quad (12)$$

where p_S denotes the probability of survival of the state $\mathbf{y}$ at the $(t+1)$ th time instant and $v_{S,t+1}(\mathbf{y}; z)$ denotes the survived component of the posterior intensity at the $(t+1)$ th time instant with the following associated mean $m_{S,t+1|t}^j$ and covariance $P_{S,t+1|t}^j$. $\gamma_{t+1}(\mathbf{y})$ denotes the birth intensity of a jammer, modeled as a predetermined Poisson RFS, with $J_{\gamma,t+1}$ number of components and associated weights $w_{\gamma,t+1}$, mean $m_{\gamma,t+1}^j$ and covariance $P_{\gamma,t+1}$. $M_{t+1|t}$ denotes the predicted number of jammers at $(t+1)$ th time instant

$$m_{S,t+1|t}^j = F m_{t|t}^j, \text{ and } P_{S,t+1|t}^j = Q_t + F P_{t|t}^j F^T$$

where $F = \begin{bmatrix} I_3 & \Delta T I_3 \\ 0_3 & I_3 \end{bmatrix}$ denotes the linear state transition matrix, ΔT denotes the update time interval of the GM-PHD filter and Q_t denotes the process noise covariance matrix. We also predict the position and velocity of our UAVs, i.e., $\mathbf{x}_{1:L,t+1|t}, \dot{\mathbf{x}}_{1:L,t+1|t}$ based on the motion model $g(\cdot, \cdot)$. Thereafter, the entire process repeats itself over time.

F. Initialization

We store the GPS signal power received at each of the receivers, p_i , $i \in \{1, \dots, L\}$ for the most recent W measurements during nonjammed, authentic conditions, i.e., before jamming is detected. The value of W is decided based on the jamming detection algorithm [50], [51] used for triggering our SLMR algorithm. After jamming is detected, we initialize our algorithm in three parts: first, we compute the average of the most recent measurements to compute $p_{i,\text{init}} = (1/W) \sum_{v=0}^{W-1} (p_{i,v-W})$, $i \in \{1, \dots, L\}$. In addition, during initialization, we also extract the position and velocity of all L UAVs and their corresponding covariance, i.e., $\mathbf{x}_{1:L,\text{init}}, \dot{\mathbf{x}}_{1:L,\text{init}}$, and $\Omega_{1:L,\text{init}}$.

Second, before we determine the actual number of jammers, i.e., $M_{\text{true},t}$, we first initialize our nonlinear GM-PHD filter by considering a stationary single jammer $M_{\text{init}} = 1$ to be present at the centroid of the known position and velocity of a network of UAVs denoted by $\mathbf{x}_{1:L,\text{init}}, \dot{\mathbf{x}}_{1:L,\text{init}}$. Thereafter, we initialize the predicted intensity $v_{0|\text{init}}(\mathbf{y})$. In addition, we also define the birth intensity $\gamma_0(\mathbf{y})$ based on the receiver and jammer characteristics. Third, we initialize our graph based on the initial constraints and measurements obtained, as explained in Section III-B.

IV. TOTAL ERROR BOUND ANALYSIS

In this section, we derive the probability of detection of multiple jammers using a network of UAVs and once detected, we estimate the error bounds of their 3-D positions. We also discuss regarding the minimum number of UAVs required for the accurate convergence of multiple jammers and UAVs.

Given that we need to detect the number of jammers and later estimate their states, the conventional notion of mean

squared error is not sufficient to characterize the total error bound. Therefore, we evaluate a more relevant metric, called OSPA [45] to compute the error characteristics, as follows:

$$e_{\text{OSPA}} = \begin{cases} \left(\frac{\lambda_1(Y_{\text{true},t}, Y_t) + \lambda_2(Y_{\text{true},t}, Y_t)}{|Y_{\text{true},t}|} \right)^{1/2}, & \text{if } |Y_t| \leq |Y_{\text{true},t}| \\ \left(\frac{\lambda_1(Y_t, Y_{\text{true},t}) + \lambda_2(Y_t, Y_{\text{true},t})}{|Y_t|} \right)^{1/2}, & \text{if } |Y_t| > |Y_{\text{true},t}| \end{cases} \quad (13)$$

where $|\cdot|$ denotes the cardinality of the RFS, $Y_{\text{true},t} = \{\beta_k, k \in \{1, \dots, M_{\text{true},t}\}\}$ represents the RFS true state, such that β_k denotes the true location and transmitted power of the k th jammer, and $|Y_{\text{true},t}| = M_{\text{true},t}$ indicates the actual number of jammers present at the t th time instant. M_t and Y_t are obtained as outputs from the measurement update of the GM-PHD filter, as seen in (9), where $|Y_t| = M_t$ represents the estimated number of jammers and $Y_t = \{\mathbf{y}_k, k \in \{1, \dots, M_t\}\}$ denotes the RFS state at the t th time instant. $\lambda_1(\cdot, \cdot)$ denotes the minimum data association error related to the 3-D positions and transmit powers of the estimated jammers as compared to the true jammers. $\lambda_2(\cdot, \cdot)$ denotes the weighted error between the estimated and actual number of jammers.

For any two RFSs U and V , the OSPA metric e_{OSPA} consisting of two components $\lambda_1(\cdot, \cdot)$ and $\lambda_2(\cdot, \cdot)$ is defined as

$$\begin{aligned} \lambda_1(U, V) &= \min_{\pi \in \Pi_{|U|}} \left(\sum_{k=1}^{|V|} d_\epsilon^2(\mathbf{u}_k, \mathbf{v}_{\pi(k)}) \right) \\ \lambda_2(U, V) &= \epsilon^2(|U| - |V|) \end{aligned} \quad (14)$$

where $\mathbf{u}_k \in U$, $\mathbf{v}_{\pi(k)} \in V$, and $\Pi_{|U|}$ denotes all the possible sets of permutations on $\{1, 2, \dots, |U|\}$ where $\pi(k)$ denotes the k th element in the set $\pi \in \Pi_{|U|}$. $d_\epsilon(\mathbf{u}, \mathbf{v}) = \min(\epsilon, d(\mathbf{u}, \mathbf{v}))$ is the distance between $\mathbf{u}$ and $\mathbf{v}$ cut off at $\epsilon > 0$. The cut-off threshold ϵ is empirically defined to capture the characteristics of the error $d(\mathbf{u}, \mathbf{v})$ while upper bounding the values of the OSPA metric. The threshold ϵ' is empirically chosen to penalize $\lambda_2(\cdot, \cdot)$ as compared to $\lambda_1(\cdot, \cdot)$.

The lower error bound e_{LB} of the OSPA metric, i.e., $e_{\text{OSPA}} \geq e_{\text{LB}}$ depends on the configuration of our UAV network, their corresponding motion model and received power. Furthermore, the error also depends on the number of jammers and their geographical separation. To evaluate the theoretical lower bound of the OSPA metric seen in (13), we compute the bounds on $\lambda_1(\cdot, \cdot)$ and $\lambda_2(\cdot, \cdot)$ separately.

As seen in (14), λ_1 , associates one estimated jammer with one true jammer such that the sum of mean squared error is minimized. Therefore, we can rewrite $\pi(k) = k$, based on which we evaluate the lower bound on $d(\beta_k, \mathbf{y}_k)$ by utilizing the PCRB [46] framework. To estimate λ_2 , we formulate the error probability via Neyman–Pearson decision rule [47].

Analogous to the Cramér Rao lower bound [60], PCRB recursively measures the maximum information that can be extracted from a nonlinear dynamical system, when motion model is stochastic. Given the time variant parameters involved, we employ the PCRB framework to evaluate the

performance of our nonlinear GM-PHD filter as

$$d(\beta_{k,t}, \mathbf{y}_{k,t}) \geq J_k^{-1} \quad (15)$$

and

$$J_k = \mathbb{E} \left[(\nabla_{\mathbf{y}_k} \ln p(z_{1:L,1:t}, \mathbf{y}_{k,1:t})) (\nabla_{\mathbf{y}_k} \ln p(z_{1:L,1:t}, \mathbf{y}_{k,1:t}))^T \right] \quad (16)$$

where J_k is the Fisher Information (FI) matrix [60], $\mathbb{E}[\cdot]$ denotes the expectation operator, $\nabla_{\mathbf{y}_k}$ denotes the gradient operator with respect to the k th jammer, and $\beta_k = [x, y, z, \phi]$ is the true location and transmit power of the k th jammer.

Using the motion model equations and the initial probability of jammer location $p(\mathbf{y}_{k,0})$, we express the joint probability of the state and measurements, denoted by $p(z_{1:L,1:t}, \mathbf{y}_{k,1:t})$ as

$$p(z_{1:L,1:t}, \mathbf{y}_{k,1:t}) = p(\mathbf{y}_{k,0}) \prod_{i=1}^t p(z_{1:L,i} | \mathbf{y}_{k,i}) \prod_{j=1}^t p(\mathbf{y}_{k,j} | \mathbf{y}_{k,j-1}). \quad (17)$$

We then recursively estimate the PCRB for each jammer by estimating the FI matrix described in (16) as follows:

$$J_{k,t+1} = D_t^{22} - D_t^{21} (J_{k,t} + D_t^{11})^{-1} D_t^{12} \quad (18)$$

where

$$\begin{aligned} D_t^{11} &= \mathbb{E} \left[-\nabla_{\mathbf{y}_{k,t}} \nabla_{\mathbf{y}_{k,t}}^T p(\mathbf{y}_{k,t+1} | \mathbf{y}_{k,t}) \right] \\ D_t^{12} &= \mathbb{E} \left[-\nabla_{\mathbf{y}_{k,t}} \nabla_{\mathbf{y}_{k,t}}^T p(\mathbf{y}_{k,t+1} | \mathbf{y}_{k,t}) \right] \\ D_t^{21} &= [D_k^{12}]^T \\ D_t^{22} &= \mathbb{E} \left[-\nabla_{\mathbf{y}_{k,t+1}} \nabla_{\mathbf{y}_{k,t+1}}^T p(\mathbf{y}_{k,t+1} | \mathbf{y}_{k,t}) \right] \\ &\quad + \mathbb{E} \left[-\nabla_{\mathbf{y}_{k,t+1}} \nabla_{\mathbf{y}_{k,t+1}}^T p(z_{k,t+1} | \mathbf{y}_{k,t+1}) \right]. \end{aligned} \quad (19)$$

By linearizing the nonlinear measurement model, we obtain

$$\begin{aligned} D_t^{11} &= F^T Q_{k,t}^{-1} F, \quad D_t^{12} = -F^T Q_{k,t}^{-1}, \quad D_t^{21} = -Q_{k,t+1}^{-1} F \\ D_t^{22} &= Q_{k,t+1}^{-1} + H_{k,t+1}^T R_{k,t+1}^{-1} H_{k,t+1}. \end{aligned} \quad (20)$$

Therefore, using (18)–(20), we obtain the FI matrix for each time instant as follows:

$$J_{k,t+1} = H_{k,t+1}^T R_{k,t+1}^{-1} H_{k,t+1} + (Q_{k,t} + F J_{k,t} F^T)^{-1}. \quad (21)$$

Next, the number of jammers detected at each time instant M_t using our SLMR algorithm is estimated during the measurement update described in (9). The error between the estimated and actual number of jammers is equivalent to the sum of misdetections and false alarms. This error occurs when the weight of the Gaussian component is less than the threshold or when the geometric separation between two jammers with respect to a network of UAVs is indistinguishable. The estimated weight of the Gaussian component depends on the noise covariance of the receiver signal power.

Based on (9), the jamming peak detected using our SLMR algorithm follows a distribution Γ_k given by

$$\begin{cases} A_0 : \Gamma_k = \mathcal{N}(0, \sigma_k^2 \mathbf{I}_4) \\ A_1 : \Gamma_k = \beta_k + \mathcal{N}(0, \sigma_k^2 \mathbf{I}_4) \end{cases} \quad (22)$$

where A_0 denotes the null hypothesis that the k th jammer is not detected whereas A_1 represents the alternative hypothesis that indicates the detection of the k th jamming source. σ_k^2 represents the receiver noise. The conditional pdf of a random variable $\alpha_k \in \Gamma_k$, where α_k and β_k are 4-D vectors representing the 3-D positions and transmit power of the k th jammer, i.e., $\alpha_k, \beta_k \in \mathbb{R}^4$ is given by

$$p_0(\alpha_k) = \prod_{r=1}^4 \frac{1}{\sqrt{2\pi\sigma_{kr}^2}} \exp\left\{-\frac{\alpha_{kr}^2}{2\sigma_{kr}^2}\right\} \quad (23)$$

$$p_1(\alpha_k) = \prod_{r=1}^4 \frac{1}{\sqrt{2\pi\sigma_{kr}^2}} \exp\left\{-\frac{(\alpha_{kr} - \beta_{kr})^2}{2\sigma_{kr}^2}\right\} \quad (24)$$

where $p_0(\cdot)$ denotes the conditional pdf under null-hypothesis H_0 and $p_1(\cdot)$ denotes the conditional pdf under the alternative hypothesis H_1 .

Thereafter, the log-likelihood ratio is computed as

$$\ln\left(\frac{p_1(\alpha_k)}{p_0(\alpha_k)}\right) = -\sum_{r=1}^4 \frac{\beta_{kr}}{\sigma_{kr}^2} \left(\alpha_{kr} - \frac{\beta_{kr}}{2}\right). \quad (25)$$

Using the Neyman–Pearson detection rule [47]

$$\delta_k(\alpha_k) = \begin{cases} 1, & \text{if } \tilde{\alpha}_k \geq \zeta_k \\ 0, & \text{if } \tilde{\alpha}_k < \zeta_k \end{cases} \quad (26)$$

where $\tilde{\alpha}_k = \sum_{r=1}^4 (\beta_{kr}/\sigma_{kr}^2) \alpha_r$ and ζ_k is the Neyman–Pearson decision threshold. Assuming the priors for null hypothesis and alternative hypothesis to be 0.5, we compute the average error probability for the k th jammer as

$$P_e(\delta_k) = \frac{P_0(\tilde{\alpha}_k \geq \zeta_k) + P_1(\tilde{\alpha}_k \geq \zeta_k)}{2}. \quad (27)$$

Therefore, by utilizing (21) and (27), which provide a lower bound on $\lambda_1(\cdot, \cdot)$ and $\lambda_2(\cdot, \cdot)$ respectively, and substituting in (13), we recursively estimate the lower bound e_{LB} of the OSPA at the t th time instant.

A. Convergence Criteria

The rate of convergence of our SLMR algorithm depends on the independent motion model of the UAVs. Each jammer is associated with four unknowns, i.e., 3-D position and transmit power of the jamming source, which are to be estimated via the GM-PHD filter. Therefore, to set up a fully constrained system in the presence of $M_{\text{true},t}$ jammers in the vicinity, we require at least $4M_{\text{true},t}$ UAVs. Under such conditions, the error function described in (10) is convex in nature. This is useful during the implementation of the SLMR algorithm for stationary UAVs. Furthermore, even if these minimum number of UAVs are not available, we can still estimate the number of jammers based on the multimodal distribution described in Section II-B and the corresponding reduced dimensional statistics regarding the jammer locations.

To reduce the requirements on the minimum number of UAVs needed for convergence, we utilize the sequential time-series of UAVs based on its motion model. Given that the noisy power measurements depend on distance from the jammer, the position of UAV should change significantly to capture

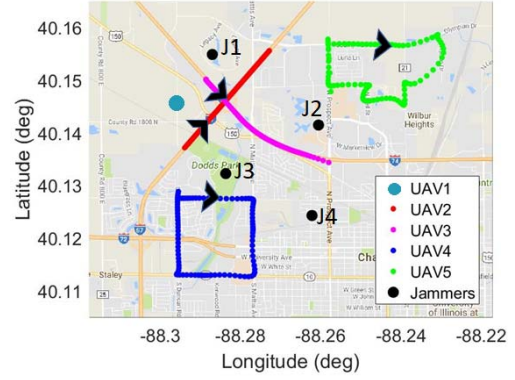


Fig. 6. Our simulated experimental setup consists of five UAVs, each equipped with a GPS receiver. The UAV trajectories are indicated by cyan, red, magenta, blue, and green. The locations of four jammers are shown in black.

the variation in received power as shown in Fig. 4. The measurements recorded by the UAV in a different position can be approximated to a new stationary UAV, but without requiring the need for additional equipment and infrastructure. Therefore, when $4M_{\text{true}} > L$, the error function, described in (11), behaves like a convex function by processing multiple time instants and converges to the global optimum over time. The convergence rate $M_t \rightarrow M_{\text{true}}$ depends on the motion of UAVs, especially the change in its position over time.

V. EXPERIMENTAL RESULTS

In this section, we validate our SLMR algorithm for various cases of simulated jamming attacks.

A. Simulated Experimental Setup

To demonstrate the performance of our SLMR algorithm, we simulated a network of 5 geographically separated UAVs, spread over a total area of 18 km, as shown in Fig. 6. We considered the motion of the UAVs to be independent of each other. Four UAVs indicated in red, blue, green and pink have uniform velocity and one of the UAVs shown in cyan is kept stationary. Also, we considered 4 stationary distributed jammers, shown by black markers in Fig. 6. We introduced simulated jamming based on the principles of linear superposition and free space path loss, as explained in Section II-A. We executed a sweep jamming attack where each jammer has the same transmit power of 50.3 W and frequency varying between $f_{L1} - 2.5$ MHz to $f_{L1} + 2.5$ MHz.

At a sampling rate of 5 MHz, we simulated the raw signal power received at each UAV and recorded the time-series of the average front-end GPS power for the simulated trajectories of all the UAVs. In the GCP-based cloud platform [23], we collectively post-processed the measurements obtained from different UAVs to simultaneously locate the multiple jammers and UAVs. We executed the GM-PHD filter at a time interval of $\Delta T = 20$ ms and the graph optimization at a rate of 0.5 s by considering the most recent $b = 25$ time instants. Later, during the execution of our algorithm, we optimize our graph using LM optimizer by employing a Cauchy kernel [61] of width 1.2 m and a damping factor of 2.2.

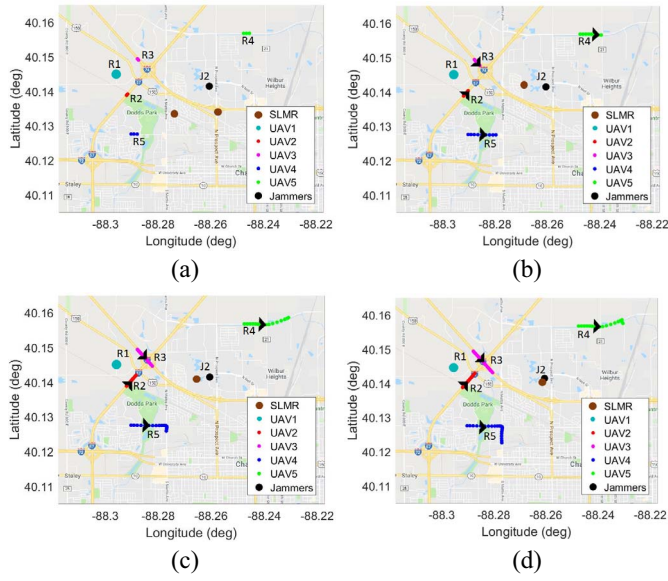


Fig. 7. Convergence rate of our SLMR algorithm to a single active $J2$ jammer. (a)–(d) Sequential snapshots of the jammer location estimated by our SLMR algorithm. The black marker denotes the actual location of the jammer while the brown marker depicts the estimate of the jammer (or jammers) at that time instant. We observed that our SLMR algorithm converges to within 4.1 m of the true jammer in ≈ 28.7 s.

B. Under the Presence of Single Jammer

In this case, we considered one active jammer $J2$ and a network of five UAVs. Fig. 7 showed the time series convergence of our SLMR algorithm to the jamming source. We observed that at time instant $t = 1.2$ s, the GM-PHD filter of our SLMR algorithm estimated two jamming sources that later converged to a single jammer. We demonstrated that our proposed SLMR algorithm estimated the true number of jammers in within 3.9 s and thereafter, locates the jammer to within 4.1 m in approximately 28.7 s.

Given that our simulated experimental setup utilizes five UAVs and we need to estimate the unknown 3-D position and transmit power of a single jammer, our system is fully constrained. Therefore, we observed the fast convergence rate of our SLMR algorithm in estimating the number of jammers. However, due to the sequential noisy power measurements, the localization error of the jammer position decreases gradually over time. In addition, utilizing the motion of the UAVs over time leads to lower errors as compared to a stationary setup of UAVs, because the geographical diversity ensures better error characterization during localization.

C. Under the Presence of Varying Number of Jammers

In this case, we evaluated the performance of the GM-PHD filter in estimating the varying number of jammers over time. As indicated by the blue line in Fig. 8, we considered all four jammers $J1$ – $J4$ mentioned in Section V-A to be active for the first 75 s and two jammers, namely $J2$ and $J4$ are disabled for the next 50 s. We observed that the estimate of the number of jammers from the GM-PHD filter, indicated by the red line, converged to the accurate number of jammers, i.e., 4 in 55 s. Similarly, after we turned off the two jammers, our PHD

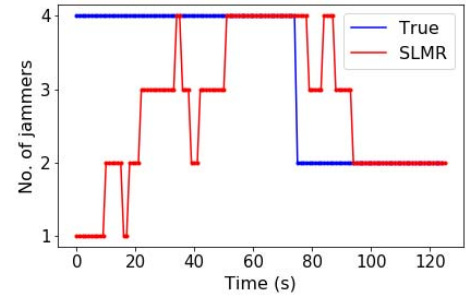


Fig. 8. Under simulated sweep continuous jamming, the number of jammers estimated using our nonlinear GM-PHD filter are indicated by the red line and the actual number by the blue line. We observed that our algorithm converged to the actual number of jammers present, i.e., 4 in ≈ 55 s and 2 in ≈ 20 s.

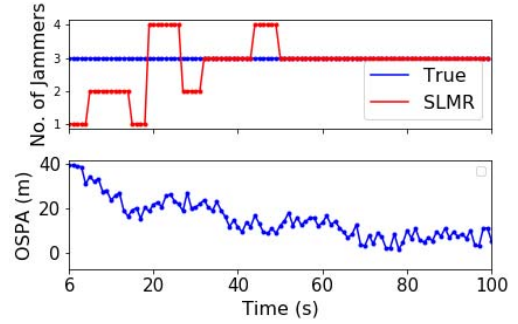


Fig. 9. Top: number of jammers estimated using our SLMR algorithm by the red line and the actual number in blue, under simulated sweep continuous jamming. We observed that our algorithm accurately converges to the actual number of jammers, i.e., 3. Bottom: convergence of the OSPA error metric across time, with 40.8 m at $t = 6$ s and decreases to 5.3 m in 64 s.

filter converged to the correct number of jammers approximately 20 s later. Due to the presence of four jammers and only five UAVs, the convergence rate for multiple jammers case is slower than the single jammer case, as seen in Section V-B.

D. Localization Accuracy of Multiple Jammers

In this case, we considered three active jammers, namely $J2$ – $J4$, as seen in Fig. 6, generating a sweep jamming.

Fig. 9(top) shows the number of jammers estimated by our SLMR algorithm, as indicated in red, whereas the blue line indicates the actual number of jammers present, i.e., 3. We observed that our algorithm efficiently utilized the motion of the UAVs and their sequential power measurements to estimate the unknown jammers in ≈ 55 s.

We initialized our SLMR algorithm by considering a single jammer present at the centroid of the UAVs and is indicated by the green dot. The convergence is depicted in Fig. 10. During $t = 0$ – 100 s, our SLMR algorithm consistently detects the $J4$ jammer, as indicated by the blue dots. By utilizing the motion of the UAVs, our algorithm detects the $J2$ jammer at time $t = 21$ s, as indicated by the orange dots. Initially, the GM-PHD filter detects the $J3$ jammer as two, indicated by the red and magenta dots, which later merge at $t = 49$ s.

Fig. 9(bottom) demonstrates the convergence of the OSPA metric, defined in (13), which indicates the total error. We

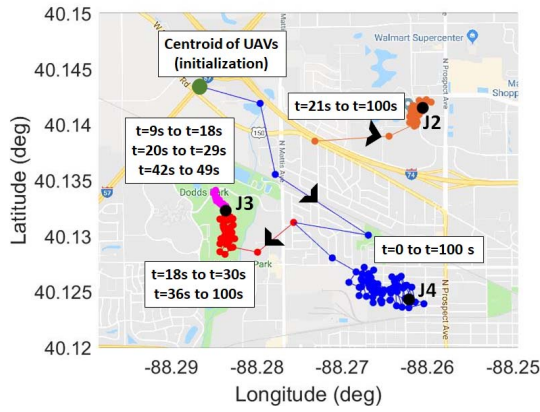


Fig. 10. Convergence of our SLMR algorithm initialized with a single jammer present at the centroid of the UAVs as indicated by the green dot. The true location of the jammers $J2$ – $J4$ are indicated by the black dots. During $t = 0$ – 100 s, our algorithm consistently detects the $J4$ jammer, as indicated by the blue dots and detects the $J2$ jammer from $t = 21$ s, as indicated by the orange dots. Our algorithm initially detects the $J3$ jammer as two different jammers, denoted by the red and magenta dots, which later merge at $t = 49$ s.

observed that the OSPA metric converged to an error of 5.3 m in ≈ 64 s. In particular, significant changes are observed until 55 s, i.e., till the number of jammers is converged and between 55 and 64 s, the jammer localization errors are further decreased by utilizing the sequential measurements processed via the GM-PHD filter and graph optimization. After convergence, we demonstrate that our algorithm accurately estimates the location of jammers to within 6-m error. This is due to the noise in the received power measurements z_i recorded at different UAVs.

E. Across Varying Jammer Transmit Power

In this case, we introduced a simulated jamming attack using $J2$ and $J4$ jammers, as seen in Fig. 6. We analyzed the performance of our SLMR algorithm for cases of 12 and 18 dB added jamming power above the noise floor for all the jammers. We separately validate that the scalar tracking algorithm fails to estimate the position of UAVs above 12 dB of added jamming. This is because with an increase in the noise floor caused by jamming, the number of satellite signals tracked fall below the minimum required number of four satellites [62]. Consequently, the navigation solution obtained via multilateration cannot be estimated. We observed that the mean position error of one jammer $J2$, as seen in Fig. 11, has an RMS error of 2.60 m and standard deviation of 4.28 m, as indicated by the blue line and an RMS error of 4.10 m and standard deviation of 5.06 m, as indicated by the red line, for 12 and 18 dB of added jamming, respectively.

Fig. 12 depicted that the position error of one UAV, denoted by $UAV4$, has an RMS error of 2.53 m and standard deviation of 3.28 m for 12 dB of added jamming and an RMS error of 3.35 m and standard deviation of 4.19 m for 18 dB of added jamming. The red line is generated for 12 dB of added jamming case while the blue line is for 18 dB of added jamming. Unlike scalar tracking, our SLMR algorithm is not sensitive to the jamming transmit power, because it utilizes these received powers as input measurements as seen in Figs. 11 and 12.

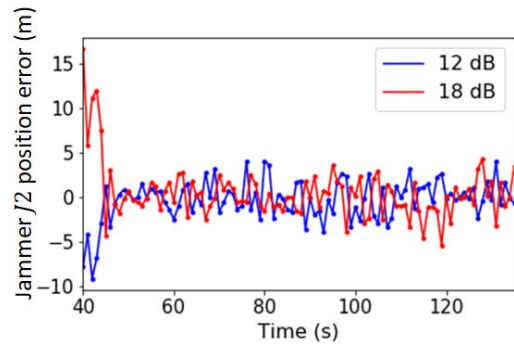


Fig. 11. Under a simulated attack, the localization error of $J2$ jammer. The red line, which depicts 12 dB added jamming power, shows an RMS error of 2.60 m and standard deviation of 4.28 m while the blue line which represents 18 dB added jamming, shows an RMS error of 4.10 m and standard deviation of 5.06 m.

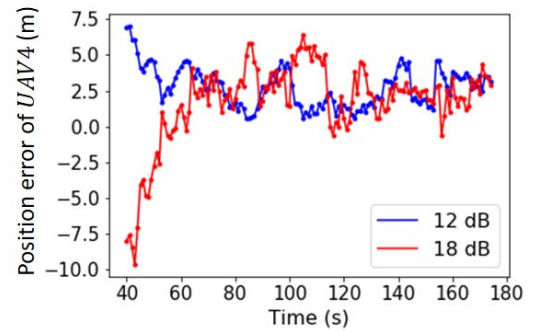


Fig. 12. Under a simulated attack, the localization error of $UAV4$. The red line, which 12 dB added jamming power, shows an RMS error of 2.53 m and standard deviation of 3.28 m while the blue line, which represents the case of 18 dB added jamming, shows an RMS error of 3.35 m and standard deviation of 4.19 m.

Unlike dead-reckoning techniques, we also demonstrated that the SLMR-based graph optimization binds the localization errors in the UAVs over time.

VI. CONCLUSION

We proposed our SLMR algorithm that is able to locate not only the multiple jammers, but also the UAVs by utilizing the received jamming signals as additional navigation sources. Using the motion model of the UAVs and the GM-PHD filter as constraints, we design a graphical framework that is optimized via an LM minimizer. We demonstrated that our SLMR algorithm accurately converges to the actual number of jammers for various cases, including attacks with a single jammer, multiple jammers and a varying number of jammers. Under the presence of single jammer, our SLMR algorithm accurately estimated the jammer in 3.9 s and localized the jammer to within 4.1 m in approximately 28.7 s. Similarly, under the presence of multiple jammers, the optimal subpattern assignment (OSPA) metric, which is used to measure the positioning accuracy of the jammers, converged to an error of 5.3 m in ≈ 64 s. In addition, unlike GPS scalar tracking, which fails to localize the UAVs under larger transmitted powers of jammers, we successfully validated that our SLMR algorithm binds the errors in the UAVs.

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